

26. What is a List in Python?

Summary:

- A list is an ordered, mutable collection.
- It can store different data types.
- Lists are created using square brackets [].

Key Point: Lists can be modified after creation.

27. What is the difference between remove(), pop(), and del?

Summary:

- remove() deletes an element by its value.
- pop() removes an element by its index and returns it.
- del deletes an element, multiple elements, or an entire list.

Key Point: Use pop() when you need the removed value.

28. What is a Tuple in Python?

Summary:

- A tuple is an ordered, immutable collection.
- Tuples are created using parentheses ().
- They are faster than lists and cannot be modified.

Key Point: Use tuples for fixed data that should not change.

29. What is a Dictionary in Python?

Summary:

- A dictionary stores data as **key-value pairs**.
- It is created using curly braces {}.
- Keys must be unique, while values can be duplicated.

Key Point: Dictionaries allow fast data lookup using keys.

30. What is the difference between a List, Tuple, and Dictionary?

Feature	List	Tuple	Dictionary
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Symbol	[]	()	{}
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Ordered	✓ Yes	✓ Yes	✓ Yes (Python 3.7+)
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Mutable	✓ Yes	✗ No	✓ Yes
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Stores	Values	Values	Key-Value Pairs
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Key Point: Choose the data structure based on your use case:

- **List** → Data changes frequently.
- **Tuple** → Data should remain constant.
- **Dictionary** → Fast access using keys.